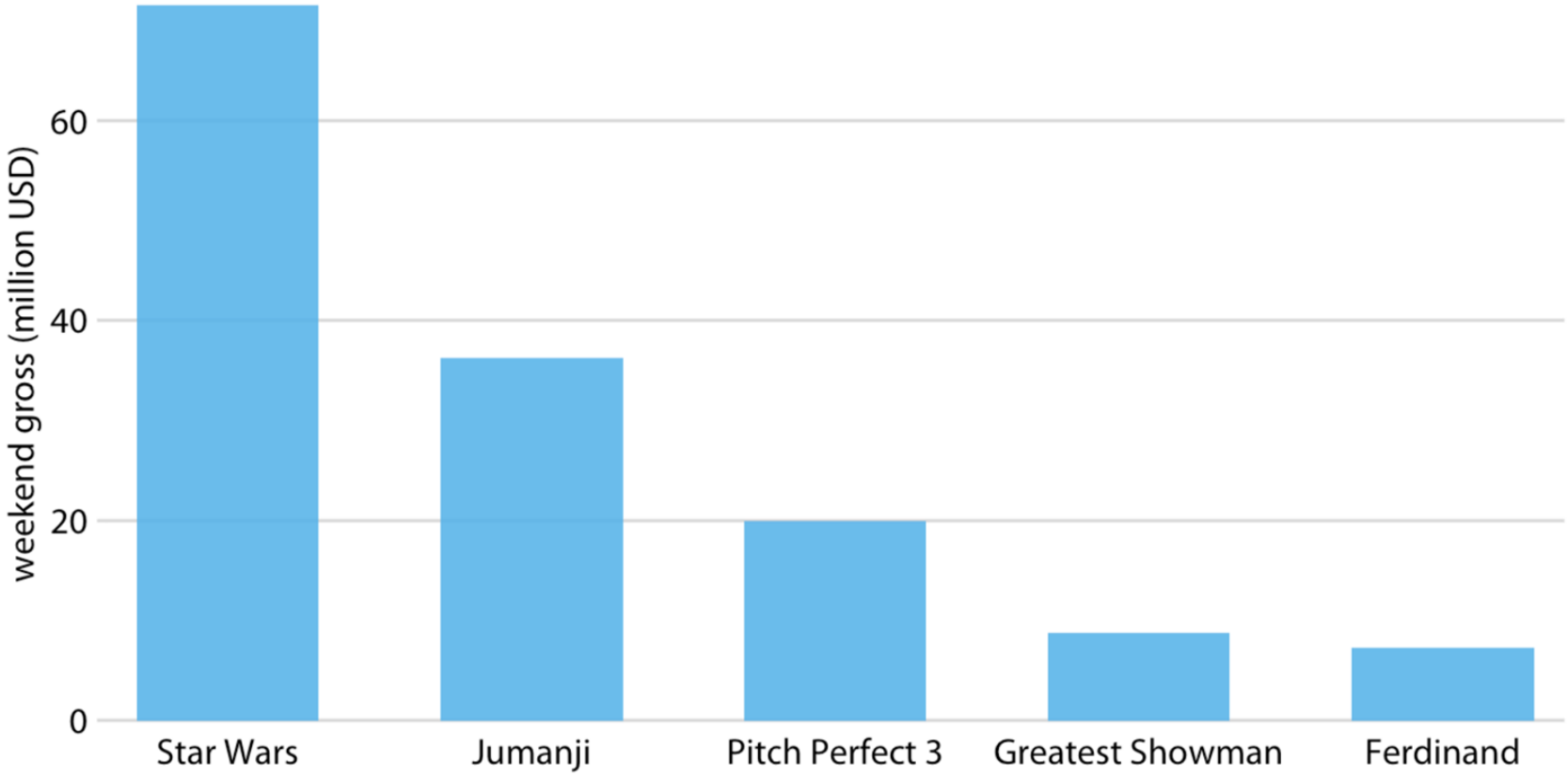
Bar Plots

These are meant for visualizing quantities associated with **categorical variables**.

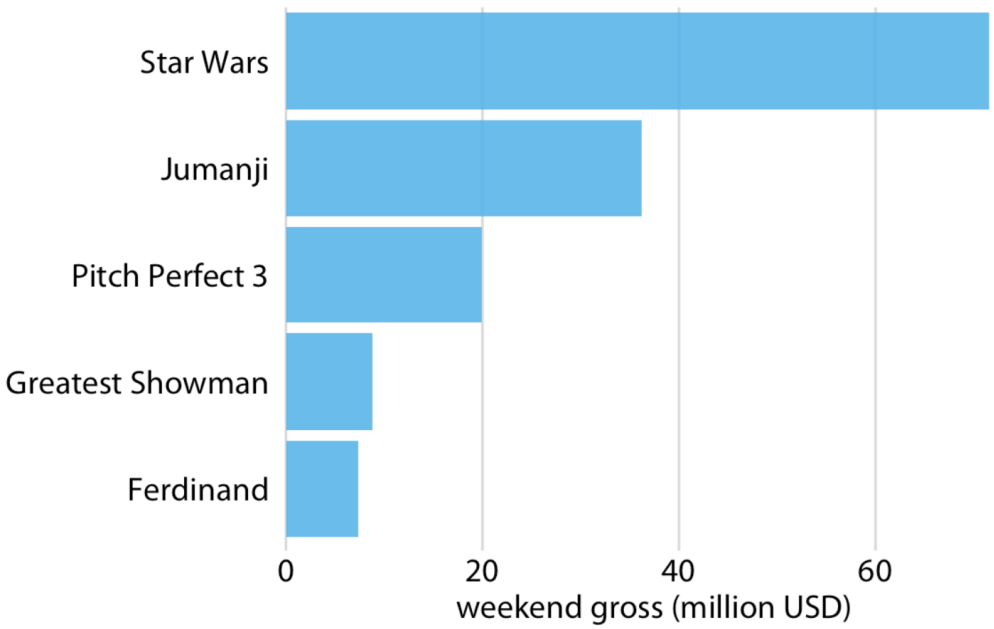


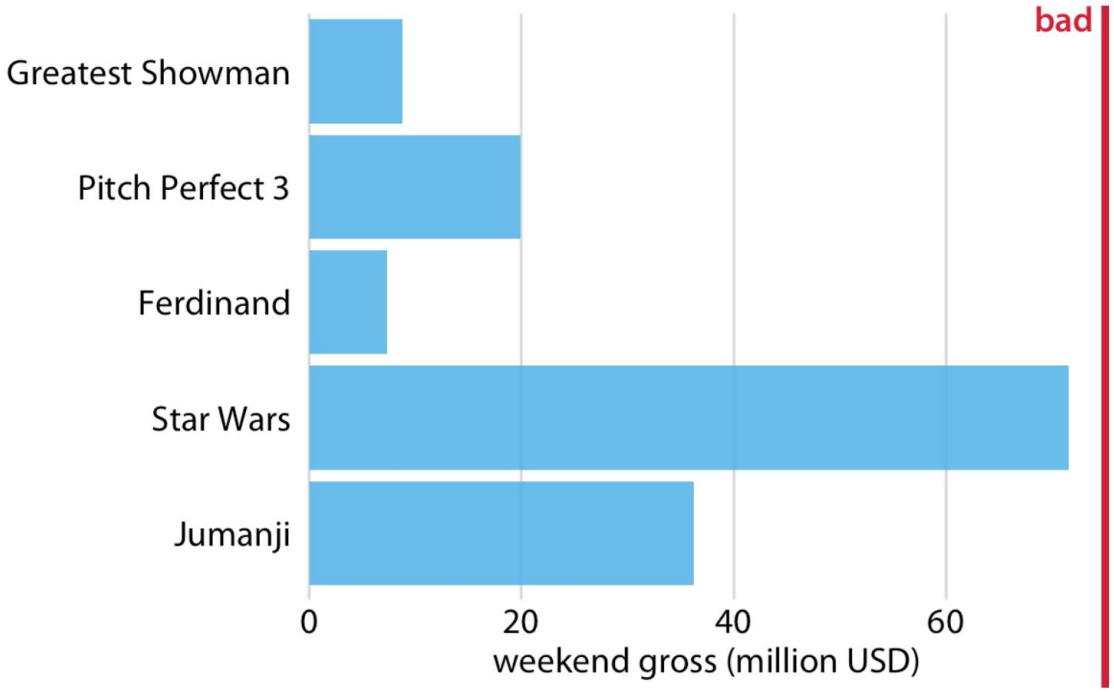
Overlapping Bar Labels: Solutions

1. If appropriate, try increasing the size of the figure.
2. If increasing size is not appropriate, create a horizontal bar plot.
3. Tilt the labels if this provides a preferred aesthetic. Generally, this is not recommended.

Bar Ordering Matters

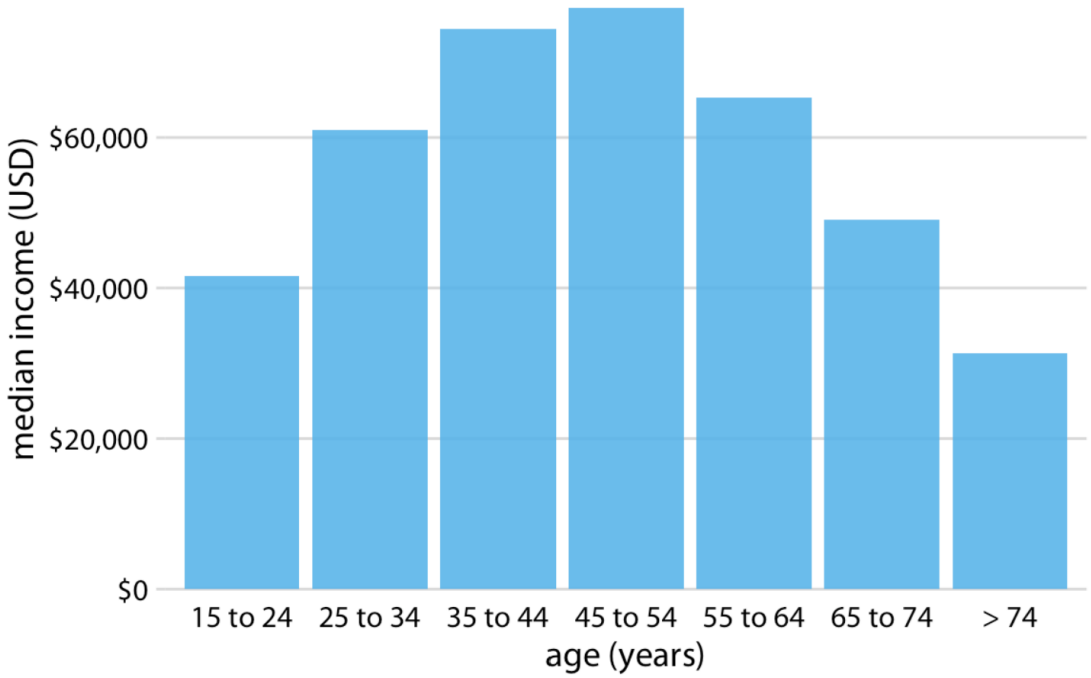
It is best to order the bars in a way that is intuitive and aligns with the information you are trying to convey.

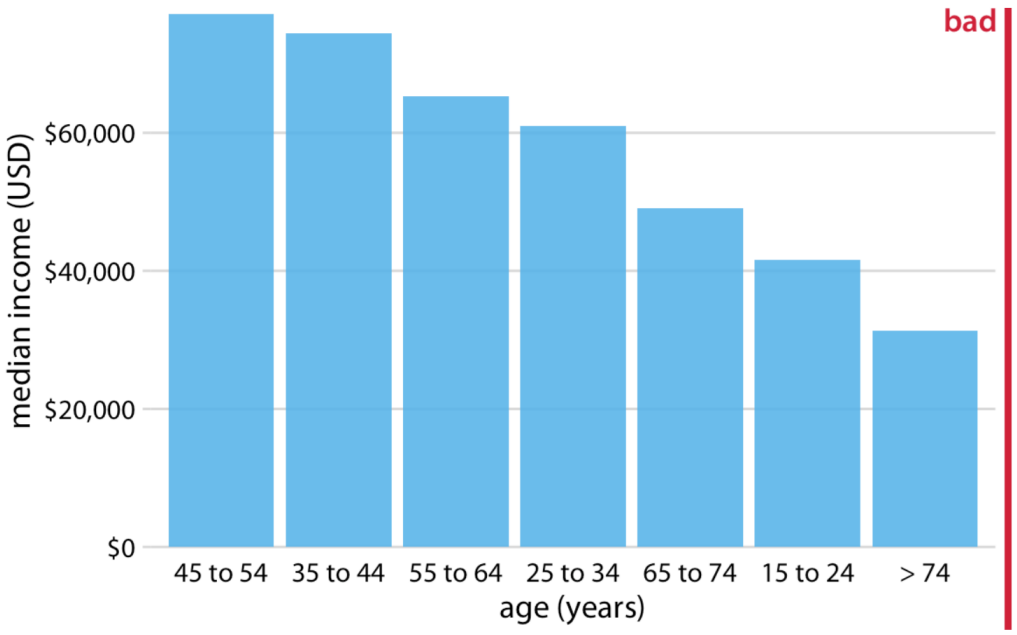




Bar Ordering Matters, Continued

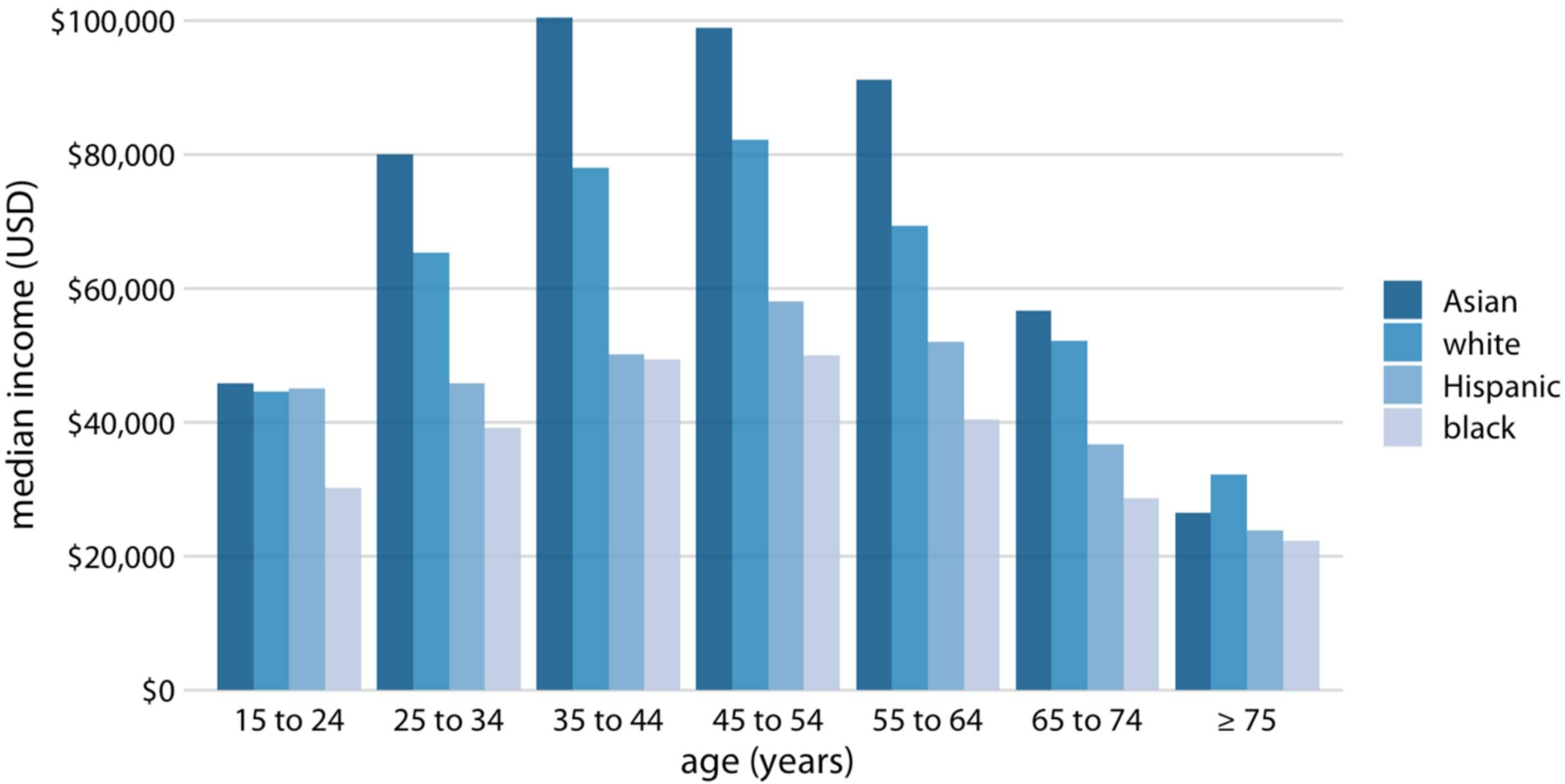
When there is a natural ordering to the levels of your categorical variable, you must maintain this ordering in the graph.

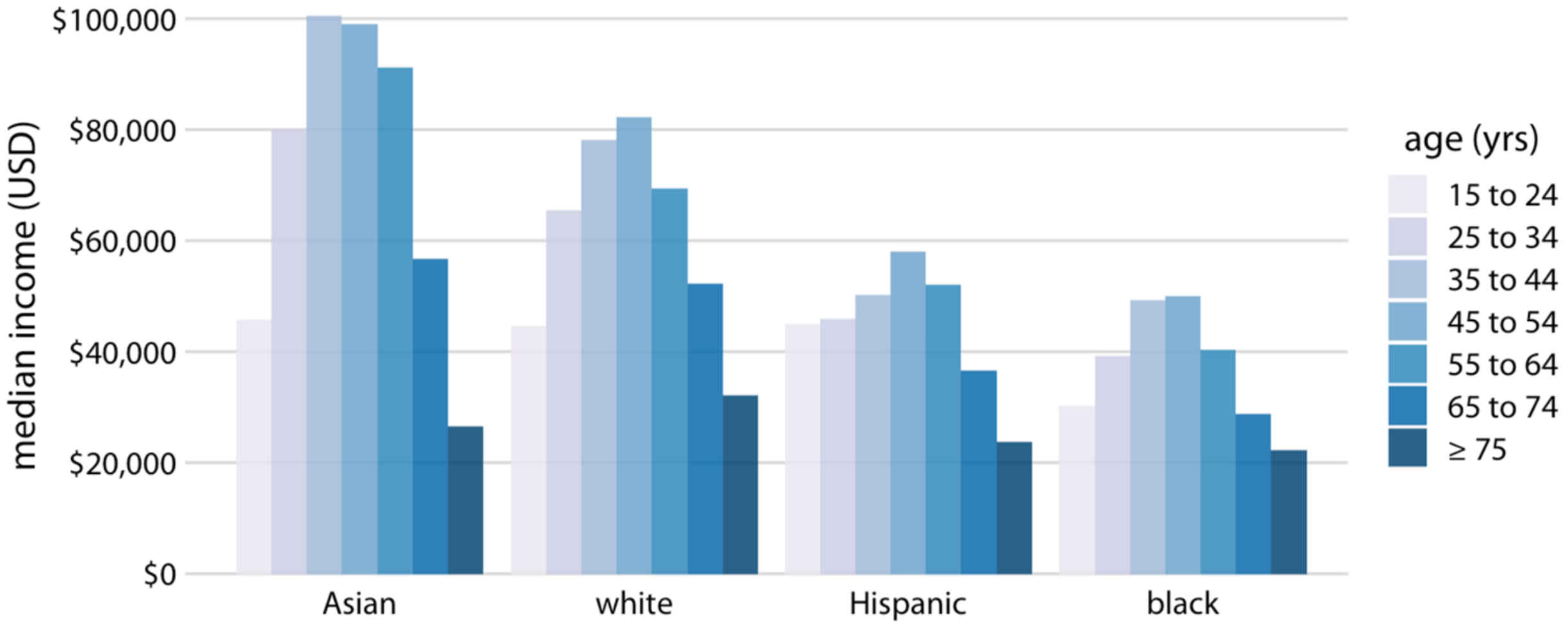


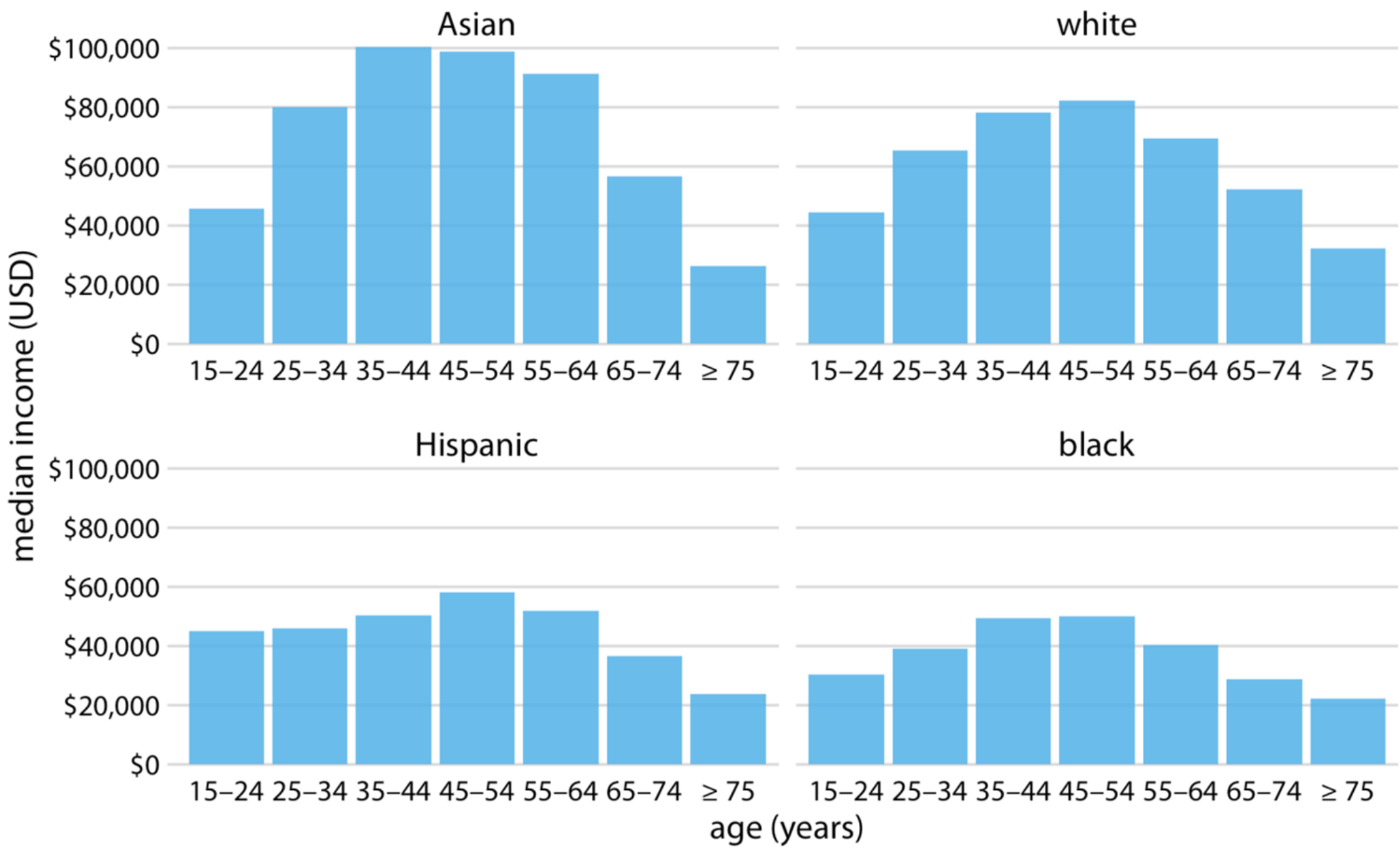


Grouped and Stacked Bar Graphs

These bar graphs are used when we want to visualize the quantities of **two** categorical variables at the same time.


$100,000
$80,000
$60,000
$40,000
$20,000
$0
median income (USD)
15 to 24
25 to 34
35 to 44
45 to 54
55 to 64
65 to 74
≥ 75
age (years)
Asian
white
Hispanic
black


$100,000
$80,000
$60,000
$40,000
$20,000
$0
median income (USD)
Asian
white
Hispanic
black
age (yrs)
15 to 24
25 to 34
35 to 44
45 to 54
55 to 64
65 to 74
≥ 75


Asian
white
Hispanic
black
$100,000
$80,000
$60,000
$40,000
$20,000
$0
median income (USD)
15–24
25–34
35–44
45–54
55–64
65–74
≥ 75
age (years)

Grouped vs. Stacked

Generally, when visualizing the quantities of two categorical variables at once, the default choice is a grouped bar graph.

Stacked bar graphs are useful when the sum of the amounts represented by the individual stacked bars is meaningful.

Ex. Total counts

Total Counts - Stacked Bar Graph Example